JAMES DE RAJA DEVARAJU

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Portfolio | LinkedIn | GitHub | XR Performance Lab

PROFESSIONAL SUMMARY

Senior Real-Time Performance Engineer with 13+ years optimizing Unity rendering pipelines, frame pacing, and **CPU/GPU** bottleneck isolation under strict **11ms/16ms** frame budgets. Specialized in XR frame timing (**72/90Hz**), **tile-based** mobile GPU architectures, and profiler-driven workflows. Optimized **200+** real-time gameplay systems and features across shipped production titles with measurable performance gains. Concurrently architected enterprise-scale API integrations and distributed systems at Zoho. Strong instrumentation mindset backed by metrics, diagnostics, and reproducible experiments. Open to remote roles and international relocation.

CORE SKILLS & TECHNOLOGIES

Rendering & Performance

- GPU/CPU Bottleneck Analysis & Classification
- Frame Pacing, Frame Timing & Budget Discipline (**11ms** / **16ms**)
- Overdraw & Fragment Pressure Reduction
- Draw Call / **SetPass** Optimization, Batching & **GPU Instancing**
- Tile-Based Mobile GPU Architecture & Memory / **GC Optimization**

XR & Profiling

- XR Stereo Rendering & Frame Timing (**72/90Hz**)
- Unity Rendering Pipelines (**URP**, Built-in) & **MSAA** Bandwidth Analysis
- **Unity 6**, **OpenXR**, **RenderDoc**, **Unity Profiler**, **Frame Debugger**
- Android Profiling, **Xcode Instruments**, Git, **C#**

Systems Engineering

- Enterprise API Integration & Distributed Systems Architecture
- Scalable Abstraction Layers, Performance Instrumentation & Diagnostics

SHIPPED TITLES & PERFORMANCE CASE STUDIES

Sneaky Warrior 3D — 3D Action / Hyper-Casual | iOS & Android

- Designed a tiered **skinned-mesh** spawn system chunking enemies into groups of 5, 10, 25, and 50 to match scene density, eliminating unnecessary draw calls and skeleton updates for off-screen or low-priority units.
- Pushed sustained FPS from sub-60 to **60+** on target mobile devices by profiling **skinned mesh** renderer cost and batching enemy cohorts to stay within **16.6ms** frame budgets.
- Applied LOD-aware spawning to reduce GPU vertex throughput and CPU animation cost during high-density combat sequences.

Snake Hole Puzzle — Worm Block Blast — Puzzle / Casual | Android

- Implemented **A*** pathfinding for real-time worm navigation across dynamic grid layouts, optimized with early-exit heuristics to keep path computation under 1ms per frame.
- Built a deterministic bullet pooling system managing **100+** concurrent projectiles with zero runtime allocations, eliminating **GC spikes** during block destruction sequences.
- Optimized block destruction VFX and physics callbacks to maintain stable **60 FPS** during peak particle and rigidbody counts on low-end Android devices.

Bolt Pop 3D — 3D Puzzle / Casual | Android

- Built a runtime **voxelisation system** to decompose arbitrary 3D objects into screw-based puzzle levels, paired with **AI-driven level generation** for rapid content authoring.
- Developed automated **constraint validation systems** to detect unsolvable regions, verify screw-to-hole count parity, and auto-resolve violations before export.
- Scaled content pipeline from manual authoring to **45+ validated levels in a single weekend**, enabling rapid iteration and publisher-ready volume.

PROFESSIONAL EXPERIENCE

Founder & Real-Time Performance Engineer2014 – Present
AlphaDen (Independent Studio) — Chennai, India

- Optimized **200+** real-time gameplay systems and features across production Unity titles under strict **16.6ms** frame budgets on **tile-based mobile GPUs**, applying profiler-driven **CPU/GPU** bottleneck classification.
- Stabilized frame pacing from unstable **10–12 FPS** to sustained **55–60 FPS** through render pipeline restructuring and bottleneck-targeted profiling.
- Built a deterministic XR benchmarking harness (**Unity 6 URP + OpenXR**) to isolate GPU fragment cost, **MSAA** bandwidth amplification, submission overhead, and frame pacing variance.
- Measured **+7.27ms GPU RenderLoop** amplification under layered transparency (overdraw stress), validated via **Unity Profiler** and **Frame Debugger** evidence.
- Evaluated XR stereo rendering cost multipliers, **MSAA** tradeoffs, and fragment scaling impact under fixed **72/90Hz** refresh targets.
- Reduced **SetPass** and draw call overhead via batching discipline, **GPU instancing**, and render state minimization.
- Eliminated runtime **GC spikes** through allocation profiling and deterministic pooling systems.
- Diagnosed masking-induced draw-call scaling in **Unity UI** rendering with **CPU/GPU** profiling and **Frame Debugger** pass inspection (**SoftMaskPro** Performance Study).

Senior Systems Engineer — Performance Optimization2017 – Present
Zoho Corporation — Chennai, India

- Architected hundreds of enterprise API integrations across major SaaS platforms, handling high-volume data exchange with reliability constraints.
- Optimized system latency and response consistency through structured instrumentation and bottleneck profiling.
- Designed scalable abstraction layers for high-concurrency distributed systems, improving maintainability and fault tolerance.
- Reduced debugging and resolution time by **23%** through structured diagnostics and standardized incident workflows.
- Decreased support incidents by **50%** via system standardization and fault isolation.
- Led architectural reviews and mentored **10+** engineers on performance-first engineering practices.

SELECTED PERFORMANCE ENGINEERING PROJECTS

XR Performance Stress Lab — Unity 6 URP + OpenXR

- Controlled **URP** benchmarking environment quantifying fragment scaling, **MSAA** cost, and submission overhead under stereo rendering constraints.
- Deterministic experiment harness producing repeatable profiler captures for **GPU RenderLoop**, overdraw, and frame pacing regression detection.

SoftMaskPro Performance Study — Unity UI Rendering

- Diagnosed and resolved masking-induced draw-call scaling through controlled scenario profiling, reducing UI rendering overhead with evidence-based optimization.
- Standardized repeatable scenario captures to validate rendering changes before and after patches.

EDUCATION

B.E., Electrical & Electronics — SSN College of Engineering

ADDITIONAL KEYWORDS

XR performance, frame pacing, frame timing, rendering optimization, overdraw, MSAA, draw calls, batching, instancing, CPU main thread, render thread, GPU RenderLoop, profiling, performance instrumentation, real-time performance optimization, mobile game development, Unity developer, C# developer, performance engineer, rendering engineer, graphics programmer